1. Problem Statement

Property management companies face significant challenges in handling maintenance requests:

- **Manual Coordination:** Requests come in via phone, email, or in-person, requiring property managers to manually classify, prioritize, and dispatch service providers.
- **Delayed Response Times:** Without automated triage, critical issues (e.g., gas leaks, flooding) may sit in the same queue as minor issues (e.g., a loose doorknob).
- **Poor Visibility:** Tenants lack real-time visibility into the status of their requests, leading to frustration and repeat contacts.
- **High Overhead:** Property managers spend excessive time on administrative tasks instead of strategic decision-making.

2. The MaintenanceAI Solution

MaintenanceAI is an AI-powered maintenance request management system that automates the entire lifecycle of a maintenance request — from submission to resolution. The system uses Google's Gemini AI to analyze natural language descriptions from tenants and automatically:

- **Classify:** categorizes the issue into one of 10 types (Plumbing, Electrical, HVAC, Structural, etc.).
- **Assess Severity:** calculates risk level dynamically (Low, Medium, High, Critical).
- **Assign Priority:** sorts urgency level from 1 (highest) to 5 (lowest).
- **Generate Remediation Plans:** provides interactive technical resolution checklists.
- **Predictive Cost:** estimates expected repair cost range for budget approvals.
- **Automated Routing:** immediately assigns the ticket to targeted service providers.

3. Target Users & System Benefits

Role	How They Use MaintenanceAI
Tenants	Submit maintenance requests in plain language. Receive instant confirmation with AI-generated ticket details and real-time status updates.
Property Managers	Use the Operations Dashboard to view, filter, assign, and manage all requests. Monitor critical issues with priority-based views.
Service Providers	Receive automated assignment notifications with issue details, severity, and recommended action steps.

4. Key Features & Integrations

4.1 Natural Language Request Submission & Multi-Modal Support

Tenants describe their issue in their own words and can upload images. The AI handles the rest, analyzing both text and attached visual evidence to get accurate context.

4.2 AI-Powered Classification & Prioritization

Using Google Gemini, the system extracts structured data from unstructured text and images, filling in the category, severity level, priority metrics, action steps, and cost estimate dynamically.

4.3 Operations Dashboard & Invoice Generation

A comprehensive, real-time dashboard for property managers featuring summary statistics cards, search and multi-filtering, detailed request logs, automated PDF invoice generation, and one-click status updates.

4.4 Automated Communication & Ticket Updates

The system automatically tracks communication between tenants and admins. When comments are posted, the system preserves history logs and automatically triggers Gemini to regenerate issue summaries.

5. AI Capabilities & Architecture Specifications

Capability	Description
Multi-Modal Analysis	Processes free-form text and images to extract structured maintenance data.
Multi-label Classification	Simultaneously determines category, severity, and priority.
Summarization	Converts informal language and history logs into professional issue summaries.
Action Recommendation	Generates step-by-step resolution plans.
Cost Estimation	Provides estimated repair cost ranges for billing.

6. Technology Stack

Layer	Technology
Frontend	Next.js 16 (App Router), React, Tailwind CSS
Backend	Next.js API Routes (server-side)
Database	SQLite via Prisma ORM
AI Engine	Google Gemini 2.0 Flash
Deployment	Fly.io (Containerized Docker deployment with persistent volumes)
Document Generation	Python (ReportLab) & jsPDF

7. Expected Business Impact

- **95% faster classification:** AI processes requests in seconds vs. minutes of manual review.
- **60% cost reduction:** Reduced administrative overhead and faster issue resolution.
- **24/7 availability:** Tenants can submit requests anytime, AI classifies immediately.
- **Improved tenant satisfaction:** Real-time visibility and faster response times.
- **Data-driven decisions:** Dashboard analytics help managers identify recurring issues.

8. Future Enhancements

- **Predictive Maintenance:** Use historical data to predict and prevent common issues.
- **Multi-language Support:** Leverage AI for automatic translation of requests.

- **Integration with IoT:** Connect smart building sensors for automatic issue detection.
- **Mobile App:** React Native companion app for tenants and service providers.